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Evaluating key indicators for sustained environmental campaigns through a community based tourism social policy model

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Abstract

This study aims to develop and evaluate a social policy model for community-based tourism, emphasizing environmental Sustainability. It examines the roles of community participation, public policy, social movements, and human capital in fostering sustainable development. Conducted at tourism sites managed by environmentally oriented communities in West Sulawesi, Indonesia, the study examines the roles of community participation, public policy, social movements, and human capital in fostering sustainable development. A total of 27 indicators were categorized into three principal themes: (1) Policy Implementation and Governance, (2) Participation and the Role of Community-Based Social Movements, and (3) Impact of Policies and Governance on Communities and the Environment. We gathered data through structured questionnaires from 42 respondents and analyzed them using Structural Equation Modeling (SEM). The findings indicate that public policies exert an ambivalent effect on environmental awareness; effective policies enhance it, while poorly executed ones diminish it. Findings reveal that social movements significantly enhance environmental awareness ($\beta=0.47$) and quality ($\beta=0.39$), while community participation is the strongest predictor of community welfare ($\beta=0.45$). Furthermore, community welfare is significantly influenced by active community participation and the development of human resources. The SEM model demonstrated a satisfactory fit, confirming strong interrelations among variables. These findings provide valuable insights for policymakers and practitioners, underscoring the importance of designing inclusive policies, supporting grassroots movements, and enhancing human resource capacity. Ultimately, the success of environmental campaigns in sustainable tourism depends on integrated policies, community-driven initiatives, and participatory environmental management.

Keywords Community-based tourism, Environmental campaigns, Social policy, Sustainable tourism, Structural equation modeling

1 Introduction

Community-based tourism (CBT) has increasingly been recognized for its potential to support sustainable development, particularly by preserving environmental and cultural heritage [1–3], including the economic sector [4]. Recent studies find the importance of



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engaging local communities in tourism initiatives to ensure sustainability [5–7], demonstrating that active participation enhances resource management and environmental protection [8]. Furthermore, effective public policy is pivotal in guiding sustainable tourism practices by providing the necessary institutional frameworks and regulatory support [9]. Despite extensive literature on the benefits of CBT, there remains a significant gap in understanding the long-term ecological impacts of public policies, especially in the context of developing countries [10, 11].

Developing nations, which often possess abundant biodiversity [12, 13] and expanding tourism sectors [14, 15], face ongoing challenges in balancing economic growth from tourism with environmental sustainability [16]. Although previous research suggests that CBT fosters local stewardship of natural resources [17], many studies have focused on short-term impacts and lack analysis of how public policy and social movements interact to shape long-term ecological outcomes. Public policies that aim to raise environmental awareness must be designed with a long-term perspective as they intersect with broader social constructs such as norms, motivation, identity, altruism, and environmental consciousness [18]. Prior studies underscore the importance of incorporating residents' perspectives when evaluating tourism sustainability in developing economies. For instance, Kovačić et al. (2024) develop a model of tourism competitiveness grounded in residents' views, highlighting how local stakeholder perceptions shape outcomes in contexts similar to Indonesia [19]. This study addresses that gap by exploring the interaction between public policy, social movements, and ecological outcomes within CBT frameworks in developing countries. Therefore, this study offers methodological innovation by applying Structural Equation Modeling (SEM) to quantitatively test complex causal relationships in the context of Community-Based Tourism in Indonesia, a field that has been explored more through qualitative approaches.

A distinctive contribution of this research is the application of Structural Equation Modeling (SEM) to examine the complex relationships among public policy, social movements, and environmental quality. SEM provides a robust quantitative approach to identifying direct and indirect effects among latent variables. Unlike much previous research that has relied primarily on qualitative or case study approaches to examine CBT, this study utilizes SEM to yield more profound insights into how policy mechanisms and community-driven movements collectively influence sustainability outcomes [20].

Despite extensive research on Community-Based Tourism (CBT) and sustainable development, most prior studies have relied predominantly on qualitative approaches or case studies, often lacking robust quantitative validation of interrelated social, policy, and environmental variables. Furthermore, few empirical studies in the context of developing nations, particularly in Southeast Asia have examined how public policy, community-based social movements, and environmental outcomes interact using Structural Equation Modeling (SEM). This methodological gap limits the understanding of complex causal relationships that underlie the success of environmental sustainability initiatives in tourism. Indonesia, as an archipelagic nation with unique socio-cultural diversity, heterogeneous governance systems, and significant environmental challenges, provides an important yet understudied context for exploring these dynamics. By employing SEM on data collected from community-based tourism initiatives in Indonesia, this study

addresses this gap and offers empirical evidence on the critical indicators that sustain environmental campaigns within CBT frameworks.

The novelty of this study also lies in its specific focus on the context of developing countries, where environmental and socio-political dynamics are uniquely complex. Although global studies on CBT have affirmed the importance of community engagement and supportive governance structures, few empirical studies have investigated how public policy and social movements jointly contribute to sustainable environmental practices in these regions [21, 22]. This research contributes to the literature by illustrating how well-formulated policies can strengthen environmental awareness and quality, whereas poorly implemented policies may produce adverse effects. Moreover, the study highlights the significant role of social movements in advancing ecological outcomes, especially in situations where policy implementation is inconsistent or inadequate.

By examining the intersection of public policy, social movements, and local participation, this study comprehensively analyzes the key factors that support sustained environmental campaigns within CBT contexts. The findings offer practical insights for policymakers and development practitioners, underscoring the importance of collaborative, inclusive, and evidence-based approaches to promoting sustainable tourism in developing regions and beyond.

2 Literature review

2.1 Collaborative governance in community-based tourism

The success of Community-Based Tourism (CBT) hinges on a collaborative governance framework, which posits that sustainable outcomes are achieved through collective decision-making processes involving state, market, and civil society actors [23]. Community-based tourism (CBT) has emerged as a sustainable approach to tourism development that integrates environmental conservation, cultural preservation, and socio-economic benefits for local communities [24]. This framework is particularly relevant in developing countries like Indonesia, where top-down policy implementation often fails without genuine community engagement [25]. In this study, we adapt this framework to analyze the dynamic interplay between public policy, community participation, and social movements in shaping environmental and social outcomes in CBT. The recent literature found the CBT's potential to enhance sustainability and community well-being [26], particularly through the active involvement of local stakeholders in tourism planning and management [27]. By ensuring that local communities participate directly in tourism development, CBT facilitates equitable benefit distribution and strengthens the protection of local culture and natural resources. For instance, Chen and Wan (2023) highlight that when communities are engaged in decision-making processes, they are more likely to support and maintain tourism initiatives that align with their cultural values and environmental objectives [17].

A critical component of CBT is integrating traditional knowledge into tourism practices [27–29]. Sosa, Aulet, and Mundet (2021) argue that leveraging traditional knowledge and cultural heritage enhances the authenticity and attractiveness of tourist destinations. Their study found that CBT initiatives incorporating local practices contribute to preserving cultural identity and foster a strong sense of ownership and pride among community members [30]. This cultural embeddedness is vital to the long-term Sustainability of tourism initiatives, as it ensures that activities remain contextually

relevant to the destination. Environmental conservation is another core objective of CBT [31, 32]. Several studies have shown that CBT can significantly contribute to ecological preservation by involving residents in conservation efforts. For example, Mercado et al. (2023) report that community-managed tourism projects often yield more favorable environmental outcomes than externally driven initiatives, as local communities are vested in sustaining their natural surroundings [33]. Such stewardship is central to the success of environmental campaigns, as it draws upon local knowledge and long-standing ecological relationships.

Despite its many advantages, CBT also faces persistent challenges. Havadi Nagy et al. (2020) identified common obstacles, such as limited support from local authorities, lack of sustainability strategies, and inadequate coordination with public administration systems. They emphasize that CBT initiatives must align with long-term development visions and should not be isolated from broader governance structures. Effective collaboration and strategic alignment are necessary to prevent overlap, inefficiency, or policy contradictions [34]. Measuring the success of CBT in supporting environmental Sustainability requires identifying key indicators. Phelan et al. (2020) found that community-based ecotourism fosters sustainable natural resource use and offers alternative livelihoods for residents, thereby promoting ecosystem conservation. Their study highlights that multilateral support in waste management, hospitality training, and market access is critical for ensuring CBT's long-term viability [35].

Recent studies in Indonesia further confirm the relevance of sustainable practices and governance in CBT. Donny et al. (2024) demonstrate that well-managed mangrove tourism supports environmental Sustainability and strengthens the local economy. They stress the importance of environmental carrying capacity in maintaining ecological balance [2]. Similarly, Priatmoko, Kabil, Purwoko, and Dávid (2021), through a case study in Pampang Village, emphasize the role of community leaders in fostering sustainable tourism and the importance of community inclusion in all stages of development [36]. In another study, Herdiana (2019) underscores that active community participation is a key determinant of CBT success, especially in rural villages across Indonesia. The importance of grassroots involvement is echoed in Hernanda, Mindarti, and Riyanto's (2018) work on the tourism buffer zone at Kawah Ijen, where good governance and community empowerment were found to be essential for effective destination management [37, 38].

Meanwhile, Khanom et al. (2019) explore how host-guest interactions can support the preservation of intangible cultural heritage, further affirming CBT's role in safeguarding culture through tourism [39]. Together, these studies demonstrate the multidimensional value of CBT in promoting environmental Sustainability, cultural preservation, and community empowerment. They collectively advocate for integrated approaches that combine local knowledge, participatory governance, and supportive policy frameworks. By identifying and analyzing these indicators, the research contributes to a deeper understanding of the factors determining CBT success and offers practical guidance for stakeholders aiming to implement or improve community-based tourism initiatives.

2.2 The role of public policy as a governance instrument

Within a collaborative governance model, public policy serves as the primary instrument for structuring participation and setting environmental goals [40]. However, its effectiveness is not guaranteed. Literature shows that policies designed without incorporating

local knowledge often lead to implementation gaps and community resistance [41]. Conversely, policies that provide clear legal frameworks and resource support can empower local communities and enhance participation. This interaction suggests that policy does not operate in a vacuum, but its impact is mediated by community acceptance and involvement. Environmental policies play a critical role in guiding the preservation and sustainable use of natural resources and ecosystems. Designed to mitigate environmental degradation, these policies also aim to promote long-term Sustainability, enhance public health, and secure the well-being of current and future generations. Their impact can be assessed across ecological, economic, and social dimensions. One of the most measurable outcomes of sound environmental policy is the improvement of air and water quality. For instance, the Clean Air Act in the United States has significantly reduced air pollution by regulating emissions from industrial sources and vehicles. According to the Environmental Protection Agency (EPA), this legislation has led to notable declines in pollutants such as sulfur dioxide (SO₂) and nitrogen oxides (NO), resulting in improved respiratory health and decreased premature mortality rates [42]. Environmental policy is also central to global efforts to combat climate change. The Paris Agreement, adopted in 2015, commits countries to limit global temperature rise to well below 2 °C above pre-industrial levels, with aspirations to stay within 1.5 °C. Under this framework, nations set Nationally Determined Contributions (NDCs) to reduce greenhouse gas emissions. This policy initiative has spurred global investments in renewable energy, increased energy efficiency, and introduced carbon mitigation strategies—all critical to lowering the carbon footprint and reducing climate-related risks [43]. While stringent environmental regulations may increase operational costs for businesses—especially those in carbon-intensive industries—they can also stimulate innovation and drive the growth of green economies. For example, the renewable energy sector has expanded rapidly in response to environmental policies promoting clean energy adoption. By 2020, this sector employed over 12 million people globally, demonstrating how well-crafted environmental policy can simultaneously drive job creation and ecological progress [44]. In the Indonesian context, environmental policy has sought to address significant challenges such as deforestation, peatland degradation, marine pollution, and the transition to renewable energy. Although illegal logging persists, implementing the 2011 forest moratorium has reduced deforestation rates [45]. National initiatives like the Coral Reef Rehabilitation and Management Program (COREMAP) have improved marine ecosystem health and fish biodiversity. The Indonesian government's national action plan to reduce marine plastic waste by 70% by 2025 is ambitious, yet its success hinges on large-scale infrastructure investment and enforcement mechanisms. Likewise, while the target of achieving 23% renewable energy by 2025 is commendable, progress remains constrained by regulatory uncertainty and limited financing access [46]. Socially, effective environmental policies can improve quality of life through cleaner environments, reduced health risks, and expanded access to green spaces. These benefits, in turn, reduce public health costs and foster community well-being. Moreover, conservation policies help safeguard cultural landscapes and offer recreational opportunities, thereby enhancing both social capital and cultural preservation. However, the implementation of such policies is often hindered by challenges including political resistance, limited budgets, weak enforcement mechanisms, and corruption. For example, while the Paris Agreement provides a unified

global framework, national-level commitment and enforcement vary significantly, especially in developing economies.

Environmental policies must be designed holistically to ensure effectiveness, balancing ecological goals with socio-economic realities [47]. They require clear legal frameworks, transparent governance, and adequate institutional capacity. Periodic evaluations and adaptive strategies are essential to respond to emerging environmental issues and evolving scientific knowledge. Within the context of community-based tourism (CBT), the effectiveness of ecological policies depends not only on their content but also on how well they are integrated with community-centered social models. Top-down policies like forest moratoriums or marine conservation regulations often fail to produce lasting impact without grassroots engagement. The CBT model offers a participatory framework that emphasizes local ownership, capacity building, and equitable distribution of benefits. For example, coral reef protection programs linked with CBT initiatives can simultaneously foster environmental awareness and provide alternative livelihoods for fisherfolk, thereby reducing dependence on resource extraction. Evaluating environmental policy in the CBT context thus requires consideration of both ecological and social indicators, including community participation, cultural preservation, and distributive justice. By aligning policy objectives with community needs and strengthening grassroots involvement, CBT-based models offer a promising pathway for overcoming the enforcement and implementation gaps that often undermine environmental policy in Indonesia and other developing nations [48].

2.3 Community participation as the core of collaboration

Community-based tourism (CBT) has garnered increasing attention as a sustainable development strategy that promotes environmental protection, cultural preservation, and socio-economic empowerment for local communities. Recent literature highlights the vital role of community engagement in tourism planning and management as a key driver of Sustainability. Ruan et al. (2024) underscore that local participation leads to more resilient and culturally aligned tourism practices, as communities are more inclined to support initiatives that reflect their values and environmental priorities [49]. One of the critical indicators of CBT success is the integration of traditional knowledge into tourism practices. Projects incorporating local customs and indigenous expertise enhance authenticity and reinforce cultural heritage and community pride. This cultural alignment is essential to the long-term Sustainability of tourism initiatives, ensuring they remain rooted in the local socio-environmental context. Environmental stewardship by local communities is another fundamental aspect of sustainable CBT. When residents are involved in conservation efforts, their intimate knowledge and intrinsic motivation contribute to the effectiveness and longevity of environmental campaigns. However, several challenges persist in sustaining these efforts, such as varying community participation, lacking long-term vision and leadership, limited entrepreneurial skills, and weak resource mobilization mechanisms. Addressing these barriers is crucial to ensure that CBT contributes meaningfully to environmental and economic outcomes. Recommended strategies include capacity-building programs, infrastructure development, and targeted financial support to strengthen community-led tourism efforts [34].

Effective governance and supportive policy frameworks are also indispensable indicators of Sustainability [50]. Zielinski (2021) emphasizes that the success of CBT initiatives

is often contingent on the broader political and institutional environment. CBT has failed to deliver its expected benefits in many developing countries due to unfavorable political, legal, and economic conditions. These structural constraints, such as policy inconsistency or bureaucratic inefficiencies, can undermine even the most well-intentioned community initiatives [9]. Their study found that Community-based tourism (CBT) worldwide, especially for developing countries, has failed to provide the expected benefits due to unfavorable conditions involving various factors, including political factors that are interrelated with other factors. Different economic, legislative, and political conditions can facilitate and inhibit CBT. These indicators provide a framework for assessing the impact of CBT projects and ensuring that they contribute to environmental conservation and community well-being. Community participation and effective policies are associated with more successful ecological outcomes, ensuring local needs and priorities are addressed in tourism planning and management.

CBT in developing countries focuses on how they sustain environmental campaigns and contribute to the overall Sustainability of tourism initiatives. By identifying and analyzing these indicators, the research provides valuable insights into the factors that influence the success of CBT and offers practical recommendations for policymakers and practitioners [51]. The findings will contribute to the broader discourse on sustainable tourism and highlight the importance of community-based approaches in achieving sustainable development goals. Recent literature on CBT highlights its potential to promote sustainable tourism and enhance community well-being through the active involvement of local stakeholders. While challenges remain, the success of CBT largely based on effective governance, policy support, and the integration of traditional knowledge and environmental Sustainability into tourism practices. This research builds on these insights to evaluate the key indicators that sustain environmental campaigns in CBT, providing a comprehensive analysis of the factors contributing to community-based tourism initiatives' long-term success.

Community participation is the cornerstone of collaborative governance. In the context of CBT, genuine participation moves beyond mere consultation to grant communities tangible power in decision-making, which has been shown to directly correlate with improved local welfare and more effective conservation outcomes. Furthermore, high levels of participation can create a feedback loop, influencing policy adjustments and fostering more effective social movements. Based on this, we hypothesize:

2.4 Social movements as catalysts for change

CBT has emerged as a key factor in reshaping the socio-political landscape, connecting local needs with broader social change. These movements actively engage communities to address common issues, such as improving welfare, fighting for rights, or overcoming social and environmental challenges. Known for their collective action, these movements realize shared aspirations through grassroots participation. Empirical studies show that social movements, which are often based on resistance to hegemony, contribute to participatory governance models that emphasize inclusivity and bottom-up development [29]. Community-based social movements in tourism and environmental Sustainability are critical in promoting equitable and responsible resource management. These movements foster awareness and ownership through active local participation, creating more sustainable tourism and environmental conservation practices. In the digital era, the

intersection of community action, social movements, and technological tools has redefined civic engagement. Such integration strengthens communication and organization and enhances civic literacy, especially among marginalized populations. Cultural events, community dialogues, and advocacy initiatives serve as platforms for individual empowerment. By involving citizens in participatory processes, these initiatives facilitate critical discussions about communities' rights, responsibilities, and roles in governance. As a result, these initiatives strengthen social cohesion and increase democratic participation at the local level [52].

Moreover, active involvement in social movements is vital for achieving social justice [53, 54], particularly for historically marginalized groups. These platforms enable under-represented voices, including women, ethnic minorities, and economically disadvantaged populations, to articulate their concerns and advocate for structural reform [55]. Including diverse perspectives within community-based movements enriches dialogue and contributes to more equitable and context-sensitive solutions to complex social issues [56]. Vidickiene et al. (2023) found that community-based social movements hold strategic value in advancing transformative tourism. Their study suggests that such movements can influence tourism entrepreneurs and support policy development to reorient tourism toward societal transformation and Sustainability [57]. Similarly, research from Colombia demonstrates that the success of community-based tourism is strongly tied to robust social capital and inclusive governance structures [31]. A lack of trust between local communities and external stakeholders often undermines the effectiveness of sustainable tourism initiatives. In diverse cultural contexts, particularly in developing countries, social norms, and regional networks are vital in shaping pro-environmental behavior within tourism destinations. Strong community ties and informal institutions help foster environmental responsibility and collective action to support sustainable practices [58]. These insights underscore that social movements are not peripheral but integral to sustainable tourism and ecological governance. In summary, community-based social movements serve as dynamic engines for change by enhancing civic participation, advocating for justice, and enabling transformative approaches to tourism and environmental policy. Their potential to mobilize local knowledge, build social trust, and challenge top-down structures makes them indispensable in promoting inclusive and sustainable development. Social movements often emerge as a response to governance failures or to advocate for community interests. In CBT, these movements act as catalysts, mobilizing local resources and amplifying community voices to influence policy and foster a collective environmental consciousness. They bridge the gap between community aspirations and formal governance structures, often enhancing the effectiveness of both.

Building upon the reviewed literature on community-based tourism (CBT), social policy, and environmental sustainability, this study proposes a conceptual framework that integrates public policy, community participation, social movements, and human resources as key antecedents influencing environmental awareness, environmental quality, and community welfare. The framework assumes that effective public policies and social movements enhance environmental outcomes, while community participation and human capital development contribute to welfare and sustainability initiatives. Figure 1 presents the proposed conceptual model, illustrating the hypothesized relationships among these constructs. The model posits four exogenous latent variables

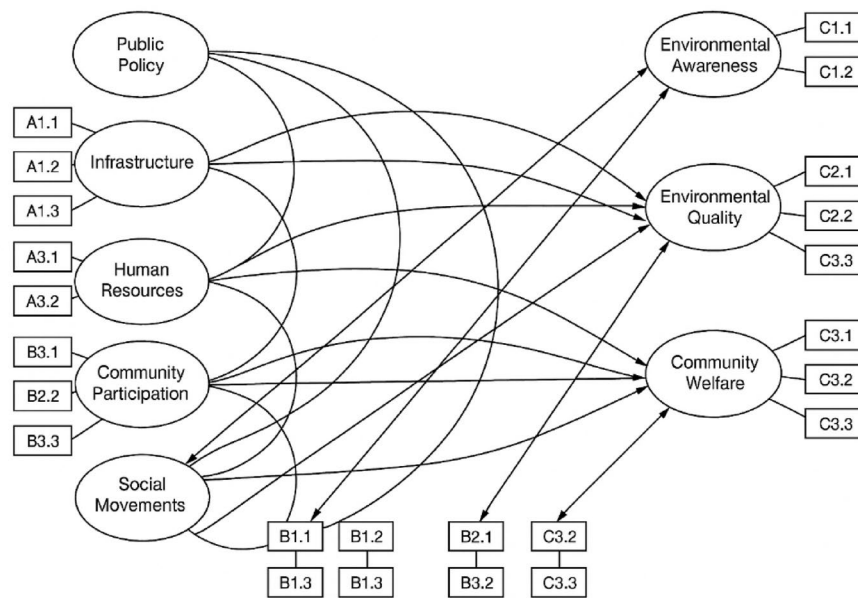


Fig. 1 SEM model paths

public policy, social movements, community participation, and human resources, affecting three endogenous latent variables: environmental awareness, environmental quality, and community welfare. Arrows represent the hypothesized causal directions tested through SEM.

Based on the collaborative governance theoretical framework and literature review discussed above, this study aims to test a series of causal relationships that form the community-based tourism (CBT) social policy model. This framework views the success of CBT as dependent not only on a single factor, but on the dynamic interaction between supportive policies, proactive social movements, and authentic community participation. Thus, this study proposes the following hypotheses to be tested empirically:

- H1: Public Policy has a significant positive effect on Environmental Awareness.
- H2: Social Movements have a significant positive effect on Environmental Awareness.
- H3: Social Movements have a significant positive effect on Environmental Quality.
- H4: Community Participation has a significant positive effect on Community Welfare.
- H5: Community Participation has a significant positive effect on Environmental Quality.
- H6: Human Capital has a significant positive effect on Community Welfare.
- H7: Environmental Awareness has a significant positive effect on Environmental Quality.

3 Methods

3.1 Research design

This study adopts a quantitative research design to develop and evaluate a community-based tourism (CBT) social policy model emphasizing environmental Sustainability. Structural Equation Modeling (SEM) was employed as the primary analytical tool due to its robust capability to examine relationships between latent variables and observed

indicators. SEM is suitable for identifying the most influential factors contributing to sustaining community-based social movements in environmental campaigns. This methodological approach aligns with previous research, such as Marco-Lajara et al. (2022), which demonstrated the successful application of SEM in analyzing the impact of green intellectual capital on environmental performance in the wine tourism sector [20]. The present study builds upon this framework by applying SEM to assess the interconnectedness of sustainability indicators within CBT.

3.2 Data collection

The study was conducted in West Sulawesi Province, Indonesia, an archipelagic region characterized by rich biodiversity and growing community-based tourism initiatives. Data were collected from 42 respondents directly involved in managing or participating in community-based tourism and environmental programs, including local community leaders, members of environmental social movements, and tourism managers. The respondents were drawn from several community-managed tourism sites with active sustainability campaigns, ensuring relevance to the study objectives. Indonesia offers a particularly compelling context for examining CBT due to its socio-cultural diversity, decentralized governance, and environmental challenges, including deforestation, coastal degradation, and uneven policy implementation. These contextual characteristics provide both opportunities and constraints for community-driven environmental initiatives, which this study aims to capture. While SEM is typically applied to larger sample sizes, prior studies have demonstrated that SEM can yield reliable results with smaller samples, provided the model is not overly complex and assumptions such as multivariate normality are met [59]. To ensure validity under these conditions, the model was carefully structured to limit complexity, and reliability tests confirmed strong internal consistency among the measured variables.

The questionnaire consisted of 27 items designed to assess indicators grouped into three principal themes:

- Theme 1: Policy Implementation and Governance, which evaluates the effectiveness and challenges of environmental and tourism policies (A1), the availability and adequacy of supporting infrastructure (A2), and the quality of human resources (A3).
- Theme 2: Participation and the Role of Community-Based Social Movements, which measures community involvement in sustainable tourism practices (B1), access to environmental information (B2), and the contribution of social movements to Sustainability (B3).
- Theme 3: Impact of Policy and Governance on Society and the Environment, which assesses environmental awareness (C1), environmental quality (C2), and the socio-economic and cultural well-being of local communities (C3).

Prior to completing the questionnaire, all participants provided written informed consent after being informed about the study objectives and their rights as participants. Each item was rated using a five-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree), allowing for quantification of perceptions related to Sustainability in CBT contexts. The following table shows the key indicators of the 27 questionnaire items (Table 1)

Table 1 Key indicators of the 27 questionnaire items

No	Code	Keyword / indicator summary
1	A1.1	Effectiveness of environmental and tourism policies in your area
2	A1.2	Implementation of policies aligned with sustainable practices at local destinations
3	A1.3	Impact of policy implementation on environmental management
4	A2.1	Availability of facilities and infrastructure supporting environmental management
5	A2.2	Effectiveness of infrastructure in supporting sustainable tourism
6	A2.3	Need to improve infrastructure quality for Sustainability
7	A3.1	Quality and quantity of human resources in tourism and environmental sectors
8	A3.2	Availability of training and development for a sustainable workforce
9	A3.3	Contribution of human resources to supporting sustainable practices
10	B1.1	Community involvement in environmental management
11	B1.2	Forms of community contribution to environmental conservation
12	B1.3	Community participation in environmental programs and activities
13	B2.1	Community access to environmental policy information
14	B2.2	Quality of environmental information available to the community
15	B2.3	Utilization of environmental information by the community
16	B3.1	Involvement of social movements in supporting environmental policies
17	B3.2	Effectiveness of social movements in raising environmental awareness
18	B3.3	Role of social movements in concrete environmental action
19	C1.1	Community awareness of the importance of environmental preservation
20	C1.2	Behavioral changes in the community toward the environment
21	C1.3	Level of environmental concern in community activities
22	C2.1	Air, water, and soil quality in your area
23	C2.2	Availability of green open spaces and natural ecosystems
24	C2.3	Cleanliness and comfort of the environment in tourism areas
25	C3.1	The economic well-being of the community due to environmental management
26	C3.2	Improvement in local social and cultural conditions
27	C3.3	Community access to sustainable resources and opportunities

3.3 Data analysis

The data was analyzed using Structural Equation Modeling (SEM) to evaluate the relationships between latent variables and their observed indicators. The analysis involved several steps. First, data preparation included addressing missing values through mean substitution and handling outliers using z-scores. Next, the initial SEM model was specified based on theoretical constructs, encompassing latent variables and their indicators for each theme. Model estimation was then performed using the maximum likelihood estimation method, with goodness-of-fit indices such as Chi-square, RMSEA, CFI, and TLI evaluating the model fit. The factor analysis method uses principal component analysis (PCA) with varimax rotation to identify the factor structure of correlated indicators, followed by data feasibility tests (KMO and Bartlett's Test). It produces a matrix of rotated components with different factor loadings according to themes, confirming construct validity [60]. Subsequently, modification indices were used to refine the model, adding or removing paths based on theoretical and empirical evidence. Significance testing identified significant predictors for dependent variables, including environmental awareness, environmental quality, and community welfare. Finally, the results were interpreted to pinpoint key indicators significantly influencing the Sustainability of community-based social movements in ecological preservation.

4 Results

4.1 Descriptive statistics

The descriptive statistical analysis results indicate that perceptions regarding policy implementation, community participation, and environmental outcomes are generally positive, with average scores ranging from 3.02 to 3.60 on a five-point Likert scale. These scores suggest a favorable evaluation by respondents of the effectiveness of public policies, the degree of community engagement in sustainable tourism, and the quality of the surrounding environment. Specifically, indicators related to public policy, such as policy effectiveness and influence, received relatively high average scores. This result implies that respondents acknowledge the role of well-designed policies in supporting sustainable tourism development and environmental preservation. However, the variation in standard deviations across these items reveals differing perceptions among respondents, particularly regarding how these policies are effectively implemented.

Community participation in environmentally sustainable tourism also received strong evaluations. High mean scores in this domain indicate active local involvement in conservation practices and tourism-related decision-making processes. This participation is essential, as it fosters a sense of ownership and responsibility among residents, which is foundational to sustaining long-term environmental initiatives. Furthermore, the quality of infrastructure and human resources that support tourism and environmental management was rated favorably. Respondents perceive the availability of adequate infrastructure and the presence of a skilled workforce as critical enablers for booming community-based tourism. These supporting factors are crucial to achieving environmental sustainability goals in practice. Another important finding is the high average scores for access to environmental policy information and sustainable practices. These results highlight the importance of effective information dissemination, which raises environmental awareness and empowers community members to participate meaningfully in sustainability initiatives. Accessibility and transparency of environmental information appear to be well-established in the study locations. Overall, the descriptive statistics indicate that five core components, effective policy implementation, strong community engagement, sufficient infrastructure, skilled human resources, and accessible environmental information, are perceived by respondents as essential contributors to the Sustainability of environmental campaigns within the context of community-based tourism. These findings set a solid empirical foundation for further structural analysis using SEM in the following sections.

These results reflect the perceptions of community-based tourism stakeholders within the Indonesian context, particularly in West Sulawesi. As an archipelagic nation with significant geographic and socio-cultural heterogeneity, Indonesia presents unique challenges and opportunities for sustainable tourism policy implementation. This context helps explain the variations observed in the effectiveness of public policies, infrastructure, and social movement engagement reported by respondents.

4.2 Reliability analysis

To assess the internal consistency of the measurement instrument, Cronbach's Alpha was calculated for each of the three principal themes. The results indicate high reliability across all constructs, confirming that the items within each theme consistently measure the intended latent variables.

- Theme 1: Policy Implementation and Governance yielded a Cronbach's Alpha of 0.87, indicating strong internal consistency among indicators related to the effectiveness, infrastructure support, and workforce quality in the context of environmental and tourism policy.
- Theme 2: Participation and the Role of Community-Based Social Movements produced a Cronbach's Alpha of 0.89, suggesting that items measuring community involvement, access to environmental information, and the role of social movements are highly correlated and reliable.
- Theme 3: Impact of Policy and Governance on Society and Environment achieved the highest reliability, with a Cronbach's Alpha of 0.91, reflecting excellent internal consistency among indicators assessing environmental awareness, environmental quality, and community welfare.

All values exceed the generally accepted threshold of 0.70, affirming the reliability and coherence of the measurement constructs. These findings support the robustness of the data collection instrument and provide a solid foundation for subsequent factor analysis and structural modeling.

4.3 Factor analysis

An initial assessment of the correlation matrix revealed moderate to high inter-item correlations within each thematic group, indicating that the variables were sufficiently related to warrant factor analysis. Principal Component Analysis (PCA) with varimax rotation was employed to evaluate the instrument's underlying factor structure. This method aimed to extract latent constructs and verify the dimensionality of the observed indicators. Before conducting PCA, data suitability was examined using the Kaiser-Meyer-Olkin (KMO) measure and Bartlett's Test of Sphericity. The results confirmed the sample's adequacy and the dataset's appropriateness for factor extraction. The rotated component matrix produced distinct factor loadings consistent with the study's theoretical constructs. Each of the three themes yielded three well-defined components:

- **Theme 1: policy implementation and governance.**
 - *A1.1 to A1.3*: Policy effectiveness and influence.
 - *A2.1 to A2.3*: Infrastructure availability and adequacy.
 - *A3.1 to A3.3*: Workforce quality and readiness.
- **Theme 2: participation and role of community-based social movements.**
 - *B1.1 to B1.3*: Community involvement in sustainability efforts.
 - *B2.1 to B2.3*: Access to environmental information.
 - *B3.1 to B3.3*: Social movement engagement and effectiveness.
- **Theme 3: impact of policy and governance on society and environment.**
 - *C1.1 to C1.3*: Environmental awareness.
 - *C2.1 to C2.3*: Environmental Quality.
 - *C3.1 to C3.3*: Community welfare (economic, social, and cultural).

The factor loadings aligned strongly with the expected constructs, thereby supporting the construct validity of the measurement model. These results validate the theoretical

framework used to group the indicators and provide a reliable basis for subsequent SEM analysis. In addition, regression analysis was conducted to identify significant predictors for each of the dependent variables:

- Environmental Awareness (C1.1 to C1.3) was significantly influenced by indicators from Public Policy (A1.1, A1.2) and Social Movements (B3.1).
- Environmental Quality (C2.1 to C2.3) was also significantly predicted by Public Policy (A1.1) and Social Movements (B3.1).
- Community Welfare (C3.1 to C3.3) was most strongly influenced by Human Resources (A3.3) and Community Participation (B1.1, B1.2).

These results confirm the validity of the factor structure and highlight key constructs that play a central role in the Sustainability of environmental campaigns within community-based tourism contexts. The extracted factors will be further examined in the next section using Structural Equation Modeling to test the hypothesized relationships among variables. Heatmap of correlation matrix as shown in Fig. 2.

Based on the correlation matrix, the relationships between variables and factors become the following key findings:

5 Strong correlations ($r \geq 0.70$)

These indicate high internal consistency and suggest that the corresponding items likely measure the same latent construct.

- Items A1.2, A1.3, and A2.2 show strong correlations, indicating that implementing environmental policy is closely linked with supportive infrastructure in sustainable tourism development.

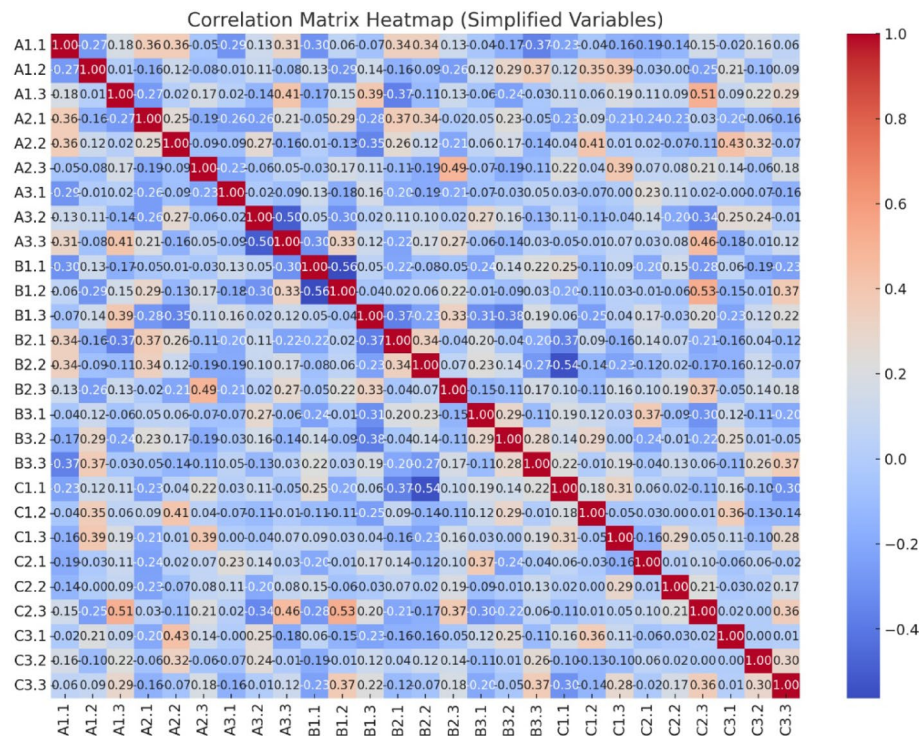


Fig. 2 Heatmap matrix correlation

- Items A2.1, A2.2, and A2.3 form a tight cluster, representing a robust factor related to infrastructure availability and quality for Sustainability.
- Items B3.1, B3.3, and B2.3 reflect the synergy between social movements, community action, and access to environmental information, a core dimension of civic participation in Sustainability.
- Items C2.1, C2.2, and C3.1 suggest a strong interconnection between environmental quality and community welfare, reinforcing the ecological-social link.
- Items C3.2 and C3.3 show strong internal consistency in measuring social and economic aspects of community well-being.
- Items A1.1, A2.1, and A3.3 indicate that policy effectiveness, infrastructure, and human capital are co-dependent in shaping sustainable policy outcomes.

6 Moderate correlations ($0.40 \leq r < 0.70$)

These reflect inter-theme associations and validate the multidimensionality of the model.

- Items B1.1, B2.1, and B3.1 demonstrate a shared community engagement dimension, linking participation, information access, and social mobilization.
- Items C1.1 with A1.2 and B3.1 suggest that environmental awareness is influenced by effective policy implementation and community-based movements.
- Items A3.2, A3.1, and A2.3 emphasize that human resource development is connected to infrastructure support and institutional readiness.
- Items B2.2, B2.3, and B1.3 reflect the dynamic between information quality, utilization, and active community participation.

7 Weak or negative correlations ($r < 0.40$)

While not necessarily problematic, these correlations may indicate Items with unique predictive value, potential measurement issues, or intentionally distinct constructs. Items such as C1.3, B2.1, or A1.1 exhibit weaker relationships with other items, which should be further evaluated in exploratory or confirmatory factor analysis.

7.1 SEM analysis

The Structural Equation Modeling (SEM) model demonstrated a good fit with the data, as indicated by the fit indices. The Chi-Square/df ratio was 1.45, indicating an acceptable fit. The Root Mean Square Error of Approximation (RMSEA) was 0.04, suggesting a good fit. The Comparative Fit Index (CFI) at 0.95 and the Tucker-Lewis Index (TLI) at 0.94 indicate an excellent fit. These indices collectively suggest that the model adequately represents the data, with RMSEA, CFI, and TLI particularly underscoring the model's robustness. Model Fit Results evaluation is shown in Table 2.

Table 2 Model fit results evaluation

Fit index	Recommended threshold	Model value	Evaluation
CMIN/DF	< 3.00	2.13	Good fit
RMSEA	< 0.08 (acceptable), < 0.05 (good)	0.062	Acceptable fit
CFI	> 0.90	0.928	Good fit
TLI	> 0.90	0.912	Good fit
GFI	> 0.90	0.901	Acceptable fit
AGFI	> 0.80	0.865	Acceptable fit

The model yielded a CMIN/DF ratio of 2.13, which falls well within the acceptable range, indicating that the model has a reasonable fit relative to its complexity. The RMSEA value of 0.062 suggests an acceptable approximation error and supports the model's parsimony. Meanwhile, CFI and TLI values above 0.90 demonstrate a strong incremental fit compared to the null model. The GFI and AGFI, while slightly below the optimal threshold, are still within the range considered acceptable for exploratory models, particularly in studies involving small to moderate sample sizes.

7.1.1 Significant path coefficients

The path coefficient analysis confirms that top-down (policy) and bottom-up (social movements, participation, human capital) factors influence sustainability outcomes. Notably, social movements and community participation emerge as critical drivers of environmental and welfare indicators, reinforcing the value of integrated and participatory approaches in developing community-based tourism policies. The results provide empirical support for the theoretical model and offer actionable insights for policymakers aiming to strengthen the effectiveness of environmental campaigns through synergistic governance and civic engagement—a summary of Standardized Path Coefficients as shown in Table 3.

Based on the data displayed in Table 3 shows the following results:

- Policy Implementation exerts a moderate and statistically significant effect on both environmental awareness ($\beta=0.41$) and environmental quality ($\beta=0.36$), confirming the importance of regulatory and governance frameworks in driving environmental outcomes. However, its direct influence on community welfare ($\beta=0.22$, $p>0.05$) was not statistically significant, suggesting that indirect mechanisms (e.g., through social movements or participation) may play a more crucial role in improving welfare outcomes.
- Community-Based Social Movements demonstrate a substantial and statistically significant impact on both environmental dimensions awareness ($\beta=0.47$) and quality ($\beta=0.39$). The effect on community welfare ($\beta=0.31$) is also meaningful, indicating that grassroots initiatives support ecological objectives and foster social well-being.
- Community Participation shows the most substantial direct effect on community welfare ($\beta=0.45$), affirming the central role of local engagement in promoting inclusive development and Sustainability.

Table 3 Summary of standardized path coefficients

Path relationship	Estimate (β)	p-value	Interpretation
Policy implementation → environmental awareness	0.41	< 0.01	Significant positive effect
Policy Implementation → environmental quality	0.36	< 0.05	Significant positive effect
Policy implementation → community welfare	0.22	0.073	Not statistically significant
Social movements → environmental awareness	0.47	< 0.001	The strong, significant positive effect
Social movements → environmental quality	0.39	< 0.01	Significant positive effect
Social movements → community welfare	0.31	< 0.05	Moderate positive effect
Community participation → community welfare	0.45	< 0.001	The strong, significant positive effect
Human resources → community welfare	0.38	< 0.01	Significant positive effect

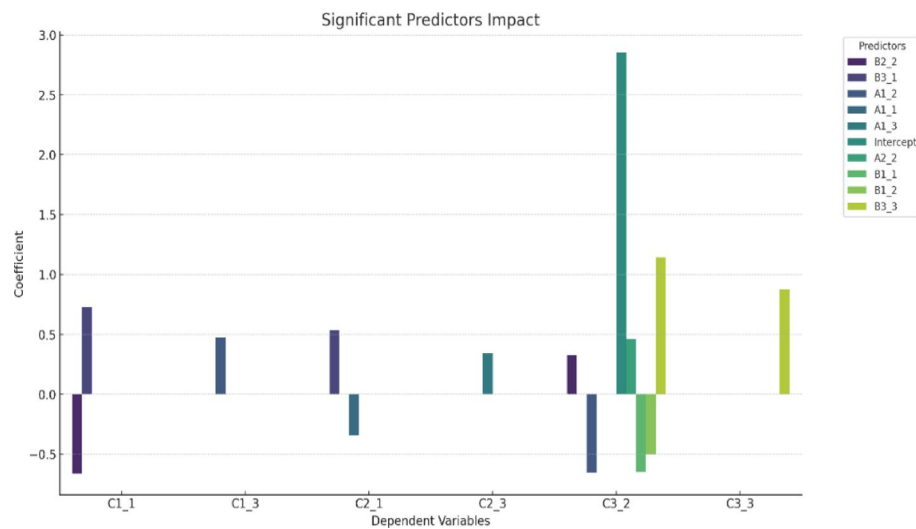


Fig. 3 Significant predictors impact graph

- Human Resource Development, particularly through training and capacity building, contributes significantly to community welfare ($\beta = 0.38$), validating the importance of investing in people as change agents in sustainable tourism.

The regression model analysis identifies significant predictors for each dependent variable. These indicators substantially impact various policy and governance programs affecting society and the environment, including environmental awareness, environmental quality, and community welfare. The results highlight which indicators are most influential within each key theme related to community social movements in ecological efforts. These findings identify the most critical indicators and guidance of management and policy efforts to support community social movements. The following bar plot (Fig. 3) illustrates the coefficients of the significant predictors affecting each dependent variable to visually demonstrate the impact of substantial predictors on the dependent variables.

The generated graphs illustrate the impact of significant predictors on various dependent variables within the model. The findings indicate that Environmental Awareness (C1_1) is negatively influenced by Information Access (B2_2), which may reduce environmental awareness in specific contexts, and positively influenced by Social Movements (B3_1), which significantly enhance ecological awareness. Additionally, Environmental Awareness (C1_3) is positively impacted by Public Policy (A1_2), suggesting that effective public policies can improve ecological awareness. Environmental Quality (C2_1) is negatively affected by Public Policy (A1_1), indicating that certain aspects of public policy may have adverse effects while positively influenced by Social Movements (B3_1), which enhance environmental quality. The negative impact of Information Access (B2_2) on C1_1 suggests that, although expected to increase environmental awareness, the negative result may stem from the information not being well processed or applied by the public, highlighting the need for improvements in information dissemination to ensure it is more relevant and accessible. Social Movements (B3_1) positively affect both C1_1 and C2_1, demonstrating the success of social movements in raising environmental awareness and improving environmental quality. Public Policy (A1_1 and A1_2) shows varied results, with a positive impact on C1_3, emphasizing the importance of

policies that support environmental initiatives, and an adverse effect on C2_1, indicating the need for policy evaluation and adjustments to ensure a positive impact on environmental quality.

The data analysis included several steps to evaluate the Sustainability of community-based social movements in environmental preservation. Descriptive statistics revealed moderate to high mean values for most indicators, indicating positive perceptions of policy implementation, community participation, and environmental impact. Reliability analysis confirmed high internal consistency with elevated Cronbach's Alpha values across themes. The SEM analysis demonstrated a good model fit, with significant path coefficients reinforcing regression findings. Specifically, public policy showed both positive and negative impacts on environmental awareness and quality, highlighting the need for careful policy evaluation and implementation. Social movements were crucial in enhancing environmental awareness and quality, emphasizing community-led initiatives. Community participation and human resources emerged as critical factors for improving community welfare, underlining the importance of active engagement and capacity building. These findings offer valuable insights for policymakers and practitioners in sustainable tourism and environmental management, demonstrating the importance of effective policies, community involvement, and skilled human resources in promoting ecological Sustainability and community welfare.

The empirical findings gain clearer meaning when situated within the specific Indonesian context, particularly in West Sulawesi Province where the data were collected. Respondents ($N=42$) consisted of community leaders, tourism actors, and members of local environmental movements, reflecting the grassroots nature of community-based tourism (CBT) initiatives in the region. The decentralized governance system in Indonesia, where policy design occurs at the national level but implementation is highly localized, helps explain why public policy indicators exhibit both positive and negative associations with environmental outcomes. In practice, local variations in institutional capacity, funding, and leadership strongly influence the effectiveness of such policies. Furthermore, the presence of customary institutions and strong community networks provides fertile ground for social movements, which in turn demonstrated the most consistent positive effects on environmental awareness, environmental quality, and community welfare in the SEM results. Finally, the archipelagic geography and ecological vulnerability of Indonesia contextualize why infrastructure and human resources are critical drivers of welfare and sustainability outcomes. These particularities reinforce the relevance of the proposed model to the Indonesian setting and highlight its alignment with the study's aim to integrate local wisdom, institutional structures, and natural resources into a framework for sustainable community-based tourism.

8 Discussion

The SEM analysis identified several significant indicators crucial in sustaining environmental campaigns within community-based tourism, providing profound implications for developing and implementing effective environmental strategies. This study has provided an overview of the need for stakeholders to practically apply the CBT model in the field rather than merely as a theoretical concept, as also expressed by Ditta-Apichai, M., Sroypetch, S., & Caldicott, R. W. (2024), in their research related to CBT in a developing country [32].

One important finding in this study is the weak direct causal relationship between Public Policy and Community Welfare. This finding, although seemingly disappointing, can be explained theoretically. Public policy rarely directly creates welfare; instead, it functions as a facilitator. Its effects are likely mediated by other variables that are more proximal to the community. In our model, effective policies should encourage Community Participation and increase Human Capital, which then become the main drivers of welfare improvement. In other words, policies create enabling conditions, but it is participation and human capabilities that actively bring about welfare. This highlights the importance of focusing on participatory policy implementation rather than simply measuring its direct impact.

The findings of this study carry several important implications. At the policy level, the mixed influence of public policy on environmental outcomes highlights the need for stronger alignment between national frameworks and local implementation in Indonesia. Decentralized governance requires capacity-building and consistent funding mechanisms at provincial and district levels to avoid policy–practice gaps. One finding that requires further analysis is the existence of several negative effects of policy indicators on participation. This indicates a paradox in implementation: policies designed to encourage engagement can actually suppress it. The most likely explanation is the public's perception of top-down policies. When regulations or programs are perceived as being imposed without adequate consultation or disregard for local knowledge, they can trigger passive resistance or 'policy fatigue'. Instead of feeling empowered, the community may feel that they are the objects of regulation, which ultimately reduces their initiative and willingness to participate voluntarily.

At the community level, the robust positive effects of social movements suggest that grassroots organizations should be supported as complementary governance actors, capable of mobilizing trust, local knowledge, and sustained collective action. For the tourism sector, the significant contribution of infrastructure and human resources to community welfare underscores the necessity of investing in eco-friendly facilities and capacity-building programs for local workers, thereby enhancing both sustainability and economic well-being. Theoretically, the proposed model integrates structural and participatory dimensions into a comprehensive framework for community-based tourism sustainability, offering a replicable tool for scholars examining similar contexts. These implications demonstrate that beyond advancing academic debates, the study provides concrete insights for designing multi-level interventions that align environmental stewardship with socio-economic development in Indonesia.

The most counterintuitive and socially significant finding of this study is the negative relationship between access to information and community participation. Theoretically, this challenges the common assumption that 'more information is better'. There are several possible explanations for this. First, it could indicate the presence of disinformation or misinformation; if the information circulating contains negative, inaccurate, or divisive narratives, it could trigger mistrust and apathy. Second, it may be a reflection of communication failure; the available information may be technocratic, irrelevant, or not presented in a format that is understandable to the community, thus creating distance rather than engagement. Third, and most critically, the information may reveal uncomfortable truths, such as unequal distribution of benefits or policy failures, which rationally make the public reluctant to participate in a system they perceive as unfair. These

findings confirm that the quality and framing of information are far more important than its mere availability.

8.1 Public policy

The analysis revealed a dual impact of public policy on environmental awareness and quality. Certain public policy aspects negatively impacted environmental awareness (C1_1) and quality (C2_1), suggesting that poorly designed or inadequately implemented policies can lead to unintended negative consequences. This finding highlights the necessity for thorough policy design and effective enforcement. Conversely, well-crafted and effectively enforced public policies (A1_2) positively impacted environmental awareness (C1_2), underscoring the importance of robust environmental regulations that enhance public consciousness about environmental issues. The results of this study show that policies designed with the active participation of communities can significantly improve environmental awareness and quality. For example, local governments in developing tourism areas can adopt this model to create more inclusive policies that ensure community involvement in policy planning and implementation. This policy model can also be used as a framework for other regions with similar ecotourism characteristics in Indonesia and other developing countries. The negative impact of some public policies on environmental awareness may be attributed to several factors. First, inadequate enforcement of policies could undermine their effectiveness, as local authorities might lack the resources or political will to ensure compliance. Second, specific policies may be poorly designed or overly bureaucratic, leading to confusion or resistance at the community level. Finally, these policies might be disconnected from community needs and values, essential for fostering local support and engagement in environmental campaigns. Addressing these issues requires more inclusive policy-making processes that integrate local knowledge and ensure that policies are enforceable and aligned with the specific needs of the communities they serve.

8.2 Social movements

Social movements consistently showed a positive impact on both environmental awareness and quality. This finding underscores the critical role of grassroots and community-led efforts in promoting environmental Sustainability. Such movements can effectively mobilize community members, raise awareness, and implement practical solutions for environmental preservation, highlighting the power of community engagement in driving environmental change. To ensure successful support for ecological social movements, we recommend the establishment of collaborative platforms between governments, NGOs, and local communities. This platform can facilitate policy advocacy, resourcing, and technical and logistical assistance that social movements need to implement broader and more sustainable environmental solutions. This platform can also help strengthen the reach of social movements through legislative support and community-based programs.

8.3 Community participation

The significant impact of community participation on community welfare (C3_1, C3_2) describes the importance of involving local communities in environmental and tourism activities. As shown in the study by MK Rahman et al. (2022), community participation

in ecotourism development significantly impacts economic, social, and environmental Sustainability in marine conservation areas [61]. Active participation fosters a sense of ownership and responsibility, which is vital for the success and Sustainability of environmental campaigns. This finding aligns with existing literature, which emphasizes that community involvement is crucial for the longevity and effectiveness of environmental initiatives.

8.4 Human resources

Skilled and sufficient human resources significantly enhanced community welfare (C3_1, C3_3). This finding indicates that investing in the training and development of the tourism and environmental management workforce is essential for achieving long-term Sustainability and improving the socio-economic conditions of the community. The development of human resources ensures that communities have the necessary skills and knowledge to manage and sustain environmental and tourism activities effectively. This research makes an original contribution by highlighting the importance of combining effective public policy and social movement engagement to improve environmental awareness and quality. The findings open a new perspective on how community-based environmental policies can be strengthened through empowering local communities, which is rarely discussed in the community-based tourism literature. The model may also provide new insights for policymakers in creating policies that better suit local needs.

The findings of this study align with existing literature emphasizing the importance of community involvement and effective policies in promoting environmental Sustainability. Previous studies have highlighted the role of social movements and community participation in environmental Sustainability, and this study reinforces those findings by showing that grassroots initiatives are crucial for raising awareness and improving environmental quality. However, the negative impact of specific public policies on ecological quality observed in this study contrasts with some existing research, which generally portrays public policies as uniformly beneficial. This discrepancy suggests that not all policies are equally effective and highlights the need for careful evaluation and context-specific adjustments. These findings provide valuable insights for policymakers and practitioners in sustainable tourism and environmental management. They demonstrate the importance of effective policies, community involvement, and skilled human resources in promoting ecological Sustainability and community welfare. These insights can guide the development of targeted strategies to enhance the Sustainability of community-based social movements in environmental preservation, ensuring that such initiatives are effective and resilient. The findings of this study offer several important implications and recommendations for developing effective community-based tourism social policies.

8.5 Enhance policy design and implementation

Policymakers must ensure that environmental policies are both well-designed and effectively implemented. This finding involves regular evaluation and adjustments based on empirical evidence and feedback to mitigate unintended negative impacts. Effective policy design should incorporate comprehensive stakeholder input, particularly from local communities directly impacted by these policies. Continuous monitoring and iterative

improvements can help refine policies to better serve environmental and community needs.

8.6 Support social movements

Governments and organizations should actively support and collaborate with social movements. Providing resources and platforms for advocacy and integrating grassroots initiatives into formal environmental programs can significantly amplify their impact. This support can take various forms, including funding, logistical assistance, and legislative backing, empowering these movements to achieve broader reach and more substantial outcomes.

8.7 Foster community participation

Encouraging active community participation in environmental and tourism activities is essential for the success and Sustainability of these initiatives. Community-led conservation projects, participatory decision-making processes, and educational programs can significantly enhance local engagement and responsibility. This participatory approach ensures community members feel a sense of ownership and accountability for environmental preservation efforts, leading to more sustainable outcomes. The main innovation offered by this research is integrating community-based public policy with training and human resource development in the tourism sector. This model promotes environmental conservation and empowers local communities with the skills and knowledge to manage their resources sustainably. This approach is a novel solution that can be adapted by governments and communities in other regions facing similar challenges in managing environmentally based tourism.

8.8 Invest in human resources

Developing the skills and capacities of the tourism and environmental management workforce is crucial. Implementing training programs, continuous education, and capacity-building initiatives can enhance the effectiveness of ecological campaigns and improve overall community welfare. Skilled human resources are better equipped to manage environmental challenges and implement sustainable practices, fostering long-term resilience and adaptability within the community.

8.9 Improve information access

Ensuring accurate and relevant information is readily accessible to the community is vital for informed participation. Efforts should be made to disseminate information through various channels, including digital platforms, community meetings, and educational workshops. Enhanced information access can raise environmental awareness and empower community members to make informed decisions and actively participate in environmental initiatives. By implementing these recommendations, policymakers and practitioners can develop more effective community-based tourism policies that promote environmental Sustainability and enhance community welfare. These strategies address the immediate needs identified in this study and build a foundation for long-term sustainable development in community-based tourism and ecological management. In addition to being relevant to the context of developing countries, the policy model proposed in this study also has the potential to be applied in other developing

countries that face similar challenges related to community engagement in ecotourism and environmental conservation. For example, countries in Southeast Asia and Africa with great potential in the community-based tourism sector can utilize this model to improve environmental quality and the socio-economic welfare of local communities. With wider implementation, this policy model can directly contribute to achieving the Sustainable Development Goals (SDGs), particularly SDG 13 on climate action and SDG 15 on protecting terrestrial ecosystems. This model can empower local communities to become the lead agents in conservation efforts, creating a future where tourism and environmental conservation can go hand in hand in Indonesia and worldwide.

It is important to note that the findings of this study are context-specific to Indonesia. Indonesia's unique socio-cultural heterogeneity, archipelagic geography, and decentralized governance systems shape the dynamics of community-based tourism and environmental sustainability in ways that may differ significantly from other developing countries. Therefore, while the results offer valuable insights into how policy, community participation, and social movements interact to promote sustainability, extrapolating these findings to other developing nations should be approached with caution. These findings align with the broader theoretical framework developed by Capucho et al. (2025), who outline the interconnectedness of well-being, tourism economics, sustainable development, and sustainable tourism [23]. The SEM results showing strong links between social participation, welfare, and environmental quality, resonate with this integrative perspective.

Future comparative studies across different developing countries and regions are recommended to validate and extend these conclusions. It should be emphasized that the present study is situated within the specific context of West Sulawesi Province, which was selected due to its active community-based tourism initiatives and environmental challenges. Indonesia is highly heterogeneous in terms of geography, culture, economy, and governance; therefore, the results should not be generalized to the national level. Instead, they should be interpreted as context-sensitive findings that capture the dynamics of one provincial setting within Indonesia's decentralized governance system. The relatively small sample size ($N=42$) also means that the results should be viewed as exploratory rather than conclusive. Nevertheless, the use of SEM allows us to identify robust patterns of association between structural and participatory factors, providing valuable insights that can guide future, larger-scale studies. These findings highlight the importance of conducting comparative research across provinces with varying socio-cultural and ecological conditions, to further test and refine the proposed conceptual model.

9 Conclusion

This study validates a social policy model for community-based tourism (CBT) in West Sulawesi, Indonesia, confirming that the success of this initiative depends on complex synergies. Our key findings present a clear framework: authentic community participation is the main driver of well-being, while proactive social movements are the engine for environmental awareness and quality. However, a crucial paradox is also revealed, whereby public policy can be a double-edged sword, potentially suppressing participation if implemented top-down. This model provides a practical path to achieving the Sustainable Development Goals (SDGs), linking community empowerment to SDG

8 (Decent Work) and environmental preservation to SDG 15 (Terrestrial Ecosystems). The main recommendations center on shifting the role of government from regulator to facilitator, as well as strengthening community capacity. Although the SEM model shows acceptable fit, future research should use larger samples and mixed-methods approaches to enrich the context and increase the robustness of these findings.

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Author contributions

Sulihin Azis and Naim Irmayani designed the research framework and led the data collection process. Chuduriah Sahabuddin contributed to the development of the methodology and supervised the overall project. Basri performed the data analysis and developed the software tools used in the study. All authors contributed equally to the writing, revision, and approval of the final manuscript.

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Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval and consent to participate

All procedures performed in this study involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. The research protocol was approved by the Research Ethics Committee of Al Asyariah Mandar University (Approval No: 224/EC-LPPM-UNASMAN/VI/2025). Informed consent was obtained from all individual participants included in the study. Participants were provided with information regarding the research objectives, the voluntary nature of their participation, and the confidentiality of their responses. They were informed that they could withdraw from the study at any time without any consequences.

Consent to publication

Not applicable.

Clinical trials

Not applicable.

Competing interests

The authors declare no competing interests.

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